Section A – Answer any 2 out of 3 questions in section A

Question 1

a) What are the key objectives for operation and design of electrical energy systems?

[3]

The key objectives are
1. security of supply (fuel security, network security),
2. cost effectiveness (support of economic growth),
3. acceptable environmental impact (local and global).

[1 mark each, 3]

b) Write the time expressions of currents and voltages for a three-phase system and derive the expressions for three phase active power and reactive power.

[6]

The expressions of currents and voltages for a three-phase system are as follows.

$v_a(t) = \sqrt{2}Vsin\omega t\ ,\ i_a(t) = \sqrt{2}Isin(\omega t - \phi)$

$v_b(t) = \sqrt{2}Vsin(\omega t - 120^o)\ ,\ i_b(t) = \sqrt{2}Isin(\omega t - \phi - 120^o)$

$v_b(t) = \sqrt{2}Vsin(\omega t - 240^o)\ ,\ i_c(t) = \sqrt{2}Isin(\omega t - \phi - 240^o)$

[1 mark each, 3]

$p_3(t) = 3VIcos\phi$

[2 marks]

$q_3(t) = 0$

[1 mark]

c) Measurement demonstrates that diversified peak demand of 100 households is 300kW, while the peak demand of each individual household is 6kW.

(i) Calculate the coincidence factor of an infinite number of household.

[3]

$P_n = j_n \times n \times P_1$

$j_n = j_\infty + \dfrac{1 - j_\infty}{\sqrt{n}}$

$j_\infty = 0.444$

[1 mark each, 3]

(ii) Briefly explain the benefits of diversification.

[1]

The main benefit of demand diversification is to reduce the total capacity of supply to meet the demand of certain number of households.

Model Answers

d) There are three identical generators, each of 90 kW capacity and availability of 98%. Assuming the reliability criterion requires that the probability to meet peak demand is above 95%, how many households [as in c)] can be supplied by these 3 generators?

[7]

The availability table of generators:

Availability (kW)	Probability (%)
270	94.1192
180	5.7624
90	0.1176
0	0.0008

[4 marks]

To meet 95% reliability requirement, the maximum peak demand can be served is 180 kW [1 Mark], which corresponds to the diversified peak demand of 58 households [2 Marks].

Most students answered a)-c) correctly. Some did wrong for (d when calculating probability or the number of households.

Model Answers

Question 2

a) What are the advantages to apply per-unit system in the analysis of power systems?

[3]

To avoid the confusion caused by:
Multiple voltage levels
Impedance of transformers depends on side
Single-phase vs three phase

[1 mark each, 3]

b) Consider a high voltage transmission circuit in Figure 1. Assume that voltages (magnitudes and phase angles) are known at both sending and receiving ends.

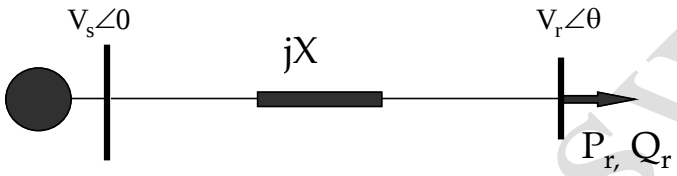


Figure 1 Transmission circuit with voltages specified at the both sending and receiving ends

(i) Write down the expressions for sending end active and reactive powers as functions of voltage magnitudes and phase angles.

[4]

$$S = V_s i_s^* = V_s \left(\frac{V_s - V_r\angle\theta}{jX}\right)^* = V_s \left(\frac{V_s - V_r\cos\theta - jV_r\sin\theta}{jX}\right)^*$$
$$= \frac{-V_s V_r \sin\theta}{X} + j\frac{V_s^2 - V_s V_r\cos\theta}{X}$$

[2 marks]

$$P_s = \frac{-V_s V_r \sin\theta}{X}$$
$$Q_s = \frac{V_s^2 - V_s V_r\cos\theta}{X}$$

[1 mark each, 2]

(ii) Based on these expressions, show that active power flow requires a difference in phase angle and that reactive power flow requires a difference in voltage magnitude.

[2]

If the angle difference θ is the same, the sin (θ) = zero, which implies that active power will be zero as well. For a small difference in angles, sin $(\theta) = \theta$, and therefore the active power is proportional to the difference in phase angles.

Model Answers

For a small difference in phase angles, $\cos(\theta) = 1$, and if the voltage magnitude difference (θ) between the receiving end and the sending end is zero, the reactive power will be zero.

[1 mark each, 2]

(iii) What is the maximum amount of active power that can be transported via this transmission line?

[2]

$P_s = \frac{V_s V_r}{X}$

For the angle difference of 90°, $\sin(\theta) = 1$.

c) There is a three-phase system as show in Figure 2. A generator is producing 45MW with 0.5PF lagging at15kV, which is fed into a 11kV substation. Please calculate the voltage (Vr) and the complex power (Sr) at the receiving end.

[9]

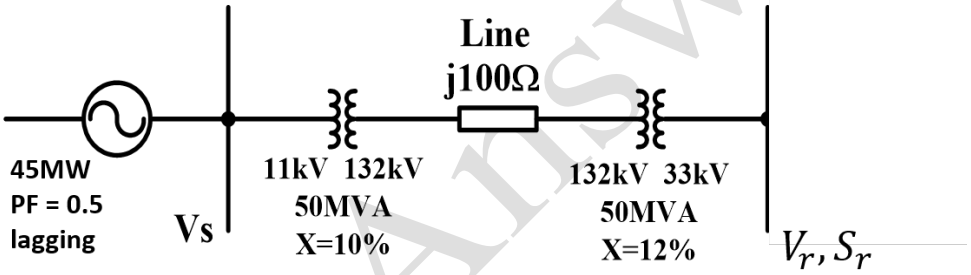


Figure 2 Three-phase system is supplied by a generator running at 45MW with 0.5PF lagging

Choose: $S_B = 50MVA;$ $V_{B11} = 11kV,\ V_{B132} = 132kV, V_{B33} = 33kV$

[1 mark]

$P_s = 0.9p.u.$

[1 mark]

$V_s = \frac{15kV}{11kV} = 1.3636\angle 0°p.u.$

[1 mark]

$\phi = \cos^{-1} 0.5 = 60°, i_s = \frac{P_s}{V_s\, PF}\angle -\phi = 1.3200\angle -60°\ p.u.$

[1 mark]

$V_r = V_s - (i_s) \times (j0.1 + j0.2870 + j0.12) = 0.8524\angle -23.11°p.u.$

[1 mark]

$|V_r| = 33kV \times 0.8524 = 28.13kV$

[1 mark]

$S_r = V_r i_r^* = 0.90 + j0.68p.u.$

Model Answers

[1 mark]

$P_r = 45\text{MW}$

[1 mark]

$Q_r = 34MVar$

[1 mark]

Most students answered a)-b) correctly. Most did wrong or didn't finish for (c when calculating P_r and Q_r in the last step

Model Answers

Question 3

A three-bus power network is presented in Figure 3 below. Data relevant for the load flow analysis on this system are given in per unit values.

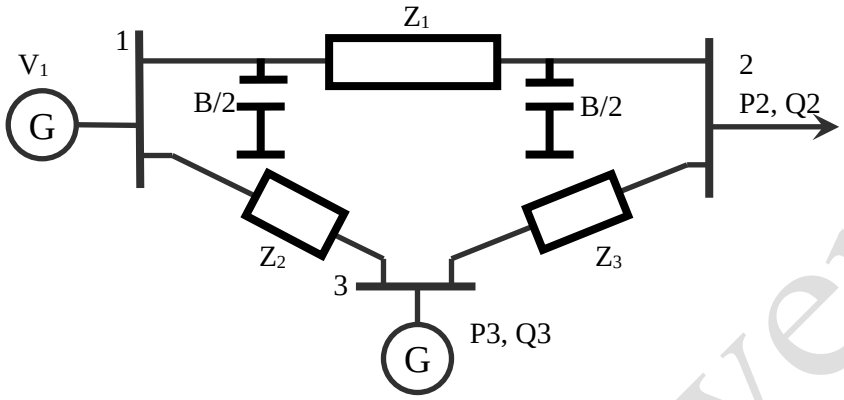


Figure 3. A three-bus power network

a) Explain the concept of the slack bus, PV bus and PQ bus in the load flow calculations

[5]

At a Slack Bus:
It serves as the voltage angle reference. As Plosses are unknown before calculation, slack bus is used to balance load and generation.

[2 marks]

At a PV bus
- P is specified
- |V| is specified
- Voltage angle is unknown
- Q is unknown
- Q adjusted to keep |V| constant
- Most generator busses are PV busses

[2 marks]

At a PQ bus
- P is specified
- Q is specified
- Voltage magnitude and angle are unknown
- Most load busses are PQ busses
-

[1 mark]

b) Given Z_1, Z_2, Z_3 and B, form the Ybus matrix

Model Answers

[6]

$$i_1 = j\frac{B}{2}V_1 + \frac{1}{Z_1}(V_1 - V_2) + \frac{1}{Z_2}(V_1 - V_3) = \left(j\frac{B}{2} + \frac{1}{Z_1} + \frac{1}{Z_2}\right)V_1 - \frac{1}{Z_1}V_2 - \frac{1}{Z_2}V_3$$
$$i_2 = j\frac{B}{2}V_2 + \frac{1}{Z_1}(V_2 - V_1) + \frac{1}{Z_3}(V_2 - V_3) = -\frac{1}{Z_1}V_1 + \left(j\frac{B}{2} + \frac{1}{Z_1} + \frac{1}{Z_3}\right)V_2 - \frac{1}{Z_3}V_3$$
$$i_3 = \frac{1}{Z_2}(V_3 - V_1) + \frac{1}{Z_3}(V_3 - V_2) = -\frac{1}{Z_2}V_1 - \frac{1}{Z_3}V_2 + \left(\frac{1}{Z_2} + \frac{1}{Z_3}\right)V_3$$

[1 mark each, 3]

$$\therefore Y = \begin{pmatrix} j\frac{B}{2} + \frac{1}{Z_1} + \frac{1}{Z_2} & -\frac{1}{Z_1} & -\frac{1}{Z_2} \\ -\frac{1}{Z_1} & j\frac{B}{2} + \frac{1}{Z_1} + \frac{1}{Z_3} & -\frac{1}{Z_3} \\ -\frac{1}{Z_2} & -\frac{1}{Z_3} & \frac{1}{Z_2} + \frac{1}{Z_3} \end{pmatrix}$$

[3 marks]

c) Assuming the per unit values $|V_1| = 1$, $|V_3| = 1.05$, $P_2 = 1$, $Q_2 = 1$, $P_3 = 1$, $Z_1 = 0.2j$, $Z_2 = 0.25j$, $Z_3 = 0.25j$ and $B = 0$, perform 2 iterations of the Gauss-Seidel load flow method.

[9]

$$V_1 = 1\angle 0° \qquad V_2^0 = 1\angle 0° \qquad V_3^0 = 1.05\angle 0°$$

[1 mark]

$$Y = \begin{pmatrix} -j9 & j5 & j4 \\ j5 & -j9 & j4 \\ j4 & j4 & -j8 \end{pmatrix}$$

[1 mark]

Iteration 1:

$$i_2 = \frac{S_2^*}{V_2^*} = Y_{21}V_1 + Y_{22}V_2 + Y_{23}V_3$$
$$V_2^1 = \frac{1}{Y_{22}}\left(\frac{S_2^*}{V_2^{0*}} - Y_{21}V_1 - Y_{23}V_3^0\right) = \frac{1}{-j9}\left(\frac{-1+j}{1} - j5 - j4 \times (1.05)\right)$$
$$= 0.9179\angle -6.9530°$$

[2 marks]

$$S_3^* = V_3^{0*}i_3 = V_3^{0*}(Y_{31}V_1 + Y_{32}V_2^1 + Y_{33}V_3^0)$$
$$= 1.05 \times (j4 + j4 \times (0.9179\angle -6.9530°) - j8 \times 1.05)$$
$$= 0.46669 - j0.7932$$
$$Q_3 = j0.7932$$
$$V_3^1 = \frac{1}{Y_{33}}\left(\frac{S_3^*}{V_3^{0*}} - Y_{31}V_1 - Y_{32}V_2^1\right) = \frac{1}{-j8}\left(\frac{1 - j0.7932}{1.05} - j4 - j4(0.9179\angle -6.9530°)\right)$$
$$= 1.05\angle 3.4602°$$

[2 marks]

Model Answers

Iteration 2:

$$V_2^2 = \frac{1}{Y_{22}}\left(\frac{S_2^*}{V_2^{1*}} - Y_{21}V_1 - Y_{23}V_3^1\right)$$
$$= \frac{1}{-j9}\left(\frac{-1+j}{0.9179\angle -6.9530°} - j5 - j4 \times (1.05\angle 3.4602°)\right)$$
$$= 0.9220\angle -6.6419°$$

[1 mark]

$$S_3^* = V_3^{1*} i_3 = V_3^{1*}(Y_{31}V_1 + Y_{32}V_2^2 + Y_{33}V_3^1)$$
$$= 1.05\angle 3.4602° \times (j4 + j4 \times (0.9220\angle -6.6419°) - j8 \times 1.05\angle 3.4602°)$$
$$= 1.0241 - j0.6970$$
$$Q_3 = j0.6970$$

$$V_3^2 = \frac{1}{Y_{33}}\left(\frac{S_3^*}{V_3^{1*}} - Y_{31}V_1 - Y_{32}V_2^2\right) = \frac{1}{-j8}\left(\frac{1 - j0.6970}{1.05\angle 3.4602°} - j4 - j4(0.9220\angle -6.6419°)\right)$$
$$= 1.05\angle 3.3043°$$

[2 marks]

Most students answered a) correctly. Some did wrong or didn't finish for (b or (c. The most common mistake is using the wrong sign for Y or S.

Model Answers

Section B – Answer any 2 out of 3 questions in section B

Question 4

a) The positive sequence sub-transient reactance of a synchronous generator is 0.10 p.u. while its positive sequence steady-state reactance is 0.01 p.u. Justify or rectify the statement with proper explanation.

[4]

The statement is incorrect as the positive sequence sub-transient reactance of a synchronous generator is less that the positive sequence steady-state reactance due to the following reason:

- Sub-transient reactance is the reactance offered by the generator during and immediately after a short-circuit during which the stator current and hence, the magnetic flux increases.
- According to the constant flux linkage theorem, flux linking a closed winding cannot change instantaneously. Therefore, the magnetic flux due to short-circuit stator current is forced through high reluctance paths that do not link the field (rotor) or damper circuits.
- Thus, the stator inductance which is inversely proportional to reluctance is initially low leading to a relatively low sub-transient reactance.
- As the flux penetrates the damper and field circuit which are lower reluctance paths, stator inductance increases resulting in a relatively high steady-state reactance.

[1 mark each]

Most students answered this correctly. Some did cover the key points but not in a logical way.

b) State the basic similarities and differences between power flow and short-circuit analyses in terms of the following:

(i) Type of the equations to be solved

[2]

- Both power-flow and short circuit analysis involves solving algebraic equations
- Power-flow equations are non-linear in terms of the solution variables (unknown voltage magnitude and phase angle) while the equations used for short-circuit analysis are linear in terms of the solution variables (short circuit voltages and currents)

[1 mark each]

Some students misinterpreted the question and drifted towards voltage vs. current balance equation

(ii) Solution technique used

Model Answers

[2]

- Non-linear power-flow equations are solved using iterative (Gauss-Seidel and/or Newton Raphson) approach
- Short-circuit analysis is solved using a direct non-iterative approach.

[1 mark each]

(iii) Matrix form of representing the transmission network

[2]

- Bus admittance matrix (Y) is used for power-flow analysis to exploit the sparsity property of Y for practical power transmission systems
- Bus impedance matrix (Z) is used for short-circuit analysis as its diagonal elements denote the Thevenin equivalent impedance at different buses which can be readily used for fault current calculation. The off-diagonal entries of Z are used to calculate the voltages at buses other than the one where short circuit occurs.

Some students failed to mention the 2nd point about bus impedance matrix

[1 mark each]

c) Why is it important to carry out short circuit analysis for planning and operation of power systems? State four important considerations.

[4]

- Determine the rating of circuit breakers which should be able to withstand and interrupt the short circuit current
- Design protection system by making sure the short circuit current is large enough to be detected
- Analyse power system stability which depends on the type and time-duration of short circuits
- Ensure power quality e.g. acceptable voltage profile which is affected by short circuits in electric proximity

[1 mark each]

Most students answered this correctly

d) A power plant has two bus bars each connected to two generators. The two bus bars are inter-connected through a reactor. Each of the four generators is rated at 15 kV, 50 MVA with a sub-transient and transient reactance of 0.15 p.u. and 0.3 p.u., respectively. At a given point, only one of the bus bars is connected to a transmission line through a 200 MVA, 15 to 275 kV transformer with 8% leakage reactance. Calculate the value of the reactance (in p.u.) needed for the reactor between the two bus bars if the three-phase short circuit level on the generator side (i.e. low voltage side) of the transformer is to be limited to 1000 MVA. Neglect the pre-fault current and fault impedance and assume the pre-fault voltage to be 1.0 p.u. Choose 100 MVA as the system base and consider fault current contribution from the generators only.

Model Answers

[6]

System base = 100 MVA

Sub-transient reactance of each generator on 100 MVA base is

$$x_G'' = 0.15 \times \frac{100}{50} = 0.3 \text{ p.u.}$$

[1 mark]

Fault-level on the generator side of the transformer to be limited to 1000/100 = 10 p.u.

[1 mark]

This corresponds to an equivalent reactance of

$$x_{eq} = \frac{1}{10} = 0.1 \text{ p.u.}$$

[1 mark]

Considering fault current contribution from the generators only the equivalent impedance looking at the fault point on the generator side of the transformer is

$$x_{eq} = \frac{x_G''}{2} \parallel \left(\frac{x_G''}{2} + x_L\right)$$

[1 mark]

Errors were mainly in this step of calculating the equivalent impedance. For instance, some students included the transformer reactance despite the fact that the fault current contribution from only the generators was supposed to be considered. Some got lost in basic series/parallel circuit simplification which was rather disappointing.

$$\Longrightarrow \frac{0.3}{2} \parallel \left(\frac{0.3}{2} + x_L\right) = 0.33$$

[1 mark]

$$\Longrightarrow x_L = 0.15 \text{ p.u.}$$

[1 mark]

Model Answers

Question 5

a) Establish the type of connection between the sequence networks for a line-to-line fault between phase B and C and show that that magnitude of the fault current (phase domain) is given by the following expression:

$$|I_f| = \frac{\sqrt{3}\,V_f}{Z_1 + Z_2 + Z_f}$$

where, Z_1, Z_2 are the equivalent positive and negative sequence impedance, Z_f is the fault impedance and V_f is the pre-fault phase voltage.

[5]

Fault currents in phase domain:
$I_A = 0, \quad I_B = -I_C$

Voltage in phase domain

$$V_B - V_C = Z_f I_B$$

[1 mark]

Fault condition in sequence domain:

$$\begin{bmatrix} I_{A0} \\ I_{A1} \\ I_{A2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}\begin{bmatrix} 0 \\ I_B \\ -I_B \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 0 \\ (\alpha - \alpha^2)I_B \\ (\alpha^2 - \alpha)I_B \end{bmatrix} = \begin{bmatrix} 0 \\ jI_B/\sqrt{3} \\ -jI_B/\sqrt{3} \end{bmatrix}$$

$I_{A1} = -I_{A2}, I_{A0} = 0$

[1 mark]

$$\begin{bmatrix} V_{A0} \\ V_{A1} \\ V_{A2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix}\begin{bmatrix} V_A \\ V_B \\ V_B - Z_f I_B \end{bmatrix}$$

$3(V_{A1} - V_{A2}) = (\alpha - \alpha^2)Z_f I_B = 3Z_f I_{A1}$
$V_{A1} - V_{A2} = Z_f I_{A1}$

[1 mark]

Some students missed the voltage equations and just assumed (without justification) a parallel connection

For LL fault, the positive and negative sequence networks are connected in parallel across the fault impedance.

$$I_{A1} = \frac{V_f}{Z_1 + Z_2 + Z_f}$$

[1 mark]

Model Answers

The fault current is given by:

$$I_B = -I_C = \frac{\sqrt{3}}{j} I_{A1} = \frac{-j\sqrt{3}}{Z_1 + Z_2 + Z_f} V_f$$

[1 mark]

A few students left it at the sequence current without reporting the phase current

b) Determine the elements of the transformation matrix (*T*) which connects the phase voltages to the c-phase components of the sequence domain voltages. Assume A-B-C and A-C-B phase sequence for positive and negative sequence components, respectively.

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = T \begin{bmatrix} V_{c0} \\ V_{c1} \\ V_{c2} \end{bmatrix}$$

[3]

$V_a = V_{a0} + V_{a1} + V_{a2} = V_{c0} + \alpha^2 V_{c1} + \alpha V_{c2}$
[1 mark]

$V_b = V_{b0} + V_{b1} + V_{b2} = V_{c0} + \alpha V_{c1} + \alpha^2 V_{c2}$
[1 mark]

$V_c = V_{c0} + V_{c1} + V_{c2}$

Hence,

$$T = \begin{bmatrix} 1 & \alpha^2 & \alpha \\ 1 & \alpha & \alpha^2 \\ 1 & 1 & 1 \end{bmatrix}$$

[1 mark]

Most students answered correctly unless there was a mistake associated with incorrect phase sequence

c) A balanced three-phase supply feeds a three-phase delta connected load which draws a line current of 250 A from the supply. Due to a sudden open circuit, the current in phase A drops to zero. Under this condition, calculate the magnitude and phase angle of the positive- (A-B-C), negative- (A-C-B) and zero-sequence components of the currents in the unaffected phases using the current in Phase B as the reference (i.e. phase angle of 0 degree).

[5]

$I_A = 0, I_B = -I_C = 250$ A

Zero sequence components

$I_{A0} = I_{B0} = I_{C0} = 0$

[1 mark]

Positive (A-B-C) sequence component

Model Answers

$$I_{A1} = \frac{1}{3}(\alpha - \alpha^2)I_B = \frac{j}{\sqrt{3}}250\text{ A} = 144.34\angle 90^o\text{ A}$$

$$I_{B1} = I_{A1} \times \alpha^2 = 144.34\angle -30^o\text{ A}$$

$$I_{C1} = I_{A1} \times \alpha = 144.34\angle 210^o\text{ A}$$

[2 marks]

Negative sequence component

$$I_{A2} = \frac{1}{3}(\alpha^2 - \alpha)I_B = -\frac{j}{\sqrt{3}}250\text{ A} = 144.34\angle -90^o\text{ A}$$

$$I_{B2} = I_{A2} \times \alpha^2 = 144.34\angle -210^o\text{ A}$$

$$I_{C2} = I_{A2} \times \alpha = 144.34\angle 30^o\text{ A}$$

[2 marks]

A few students forgot to calculate the B and C phase current of the sequence components

d) For a cylindrical rotor synchronous machine justify the following statements about steady-state and sub-transient values of sequence reactance with proper reasoning:

(i) The steady-state positive sequence reactance is larger than the steady-state negative sequence reactance

[2]

- Magnetic field produced by the positive sequence current rotates at synchronous speed in the same direction as the rotor which implies high flux linkage and hence relatively high positive sequence reactance (X_1) in steady state
- Magnetic field produced by negative sequence current rotates at synchronous speed but opposite to the direction of rotation of the rotor. Hence the induced current prevents flux from penetrating the rotor resulting in less flux linkage and hence $X_2 < X_1$

[1 mark each]

(ii) Both positive and negative sequence steady-state reactance are larger than the zero sequence steady-state reactance

[2]

- Net magnetic field produced by the zero sequence currents in theoretically zero i.e. X_0 should ideally be zero.
- However, due to leakage flux, end turns harmonic flux etc. X_0 has a small value which is usually much less than X_1 and X_2

Model Answers

[1 mark each]

(iii) Sub-transient reactance is similar for both positive and negative sequence

[1]

- Excess magnetic flux (due to short circuit) under sub-transient condition flows through similar high reluctance path outside the field and damper windings irrespective of the direction of rotating magnetic field with respect to the rotor

Most students answered d(i) correctly but the answers to part (ii) and (iii) were either incomplete or wrong in several cases

e) For a three-phase transformer with delta-star connection, explain why zero sequence current can circulate within the delta connected primary winding only if the star connected winding has its neutral grounded.

[2]

- Zero sequence current can flow in the primary winding of a transformer only if a balancing zero sequence current can flow in the secondary

- Zero sequence current can flow in a star connection only if the neutral is grounded allowing a matching zero sequence current to circulate within the phases of the delta winding

[1 mark each]

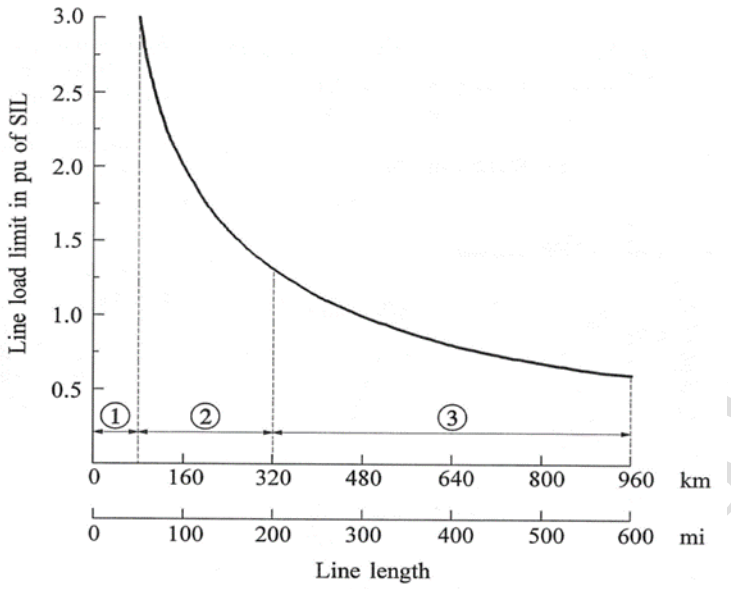
Most students answered this correctly

Model Answers

Question 6

a) The power transfer capability of an uncompensated transmission line reduces with increase in distance of transmission. Draw a typical line loadability curve covering a range of transmission distance from 0 to 1000 km and explain the factors that limit the allowable power transfer for different range of distances.

[4]



① 0–80 km: Region of thermal limitation
② 80–320 km: Region of voltage drop limitation
③ 320–960 km: Region of small-signal (steady-state) stability limitation

- Up to about 80-100 kms the transmission lines can be used up to their thermal limits.
- From 80-100 kms up to about 300-320 kms (medium lines) power transfer capacity is limited by voltage drop consideration due to large line reactance
- Beyond 320 kms (long lines) the angle stability limit becomes the dominant factor.

[2 marks each for the diagram and constraints]

Apart from some improper use of phrases (e.g. heat transfer limit etc.) most students covered the main points

b) Synchronous generators inherently provide inertial response unlike their converter-interfaced asynchronous counterparts. How is inertial response of synchronous generators helpful in reducing the drop in grid frequency?

[2]

- Drop of grid frequency is associated with decrease in speed of rotation of the synchronous generators which could be caused by an increase in load or decrease of mechanical power input.
- As generator rotor speed decreases, stored kinetic energy is released as electrical power output (known as inertial response) which reduces the extent of frequency drop.

Model Answers

[1 mark each]
The students mostly mentioned the first point but not the second one.

c) The inertia constant (H) of a thermal power generator is 5 seconds. What does this statement physically signify?

[2]

The statement signify that the generator can sustain its rated power output for 5 seconds solely on the basis of the kinetic energy stored in the rotor at synchronous speed.

d) A cylindrical rotor synchronous generator is connected to a large power system (which could be assumed to be an infinite bus) through a transmission corridor with an equivalent reactance of 0.5 p.u. The generator operates at 40% of its steady state stability limit under normal condition. The voltages at the generator terminal and infinite bus are maintained at 1.0 p.u. The resistance of the generator and the transmission corridor is negligible.

(i) The mechanical power input to the generator is increased by 20%. Use the equal area criterion to show that the maximum excursion of the generator rotor angle (δ_{max}) can be obtained from the following equation assuming a constant generator terminal voltage.

$$0.48 \times \delta_{max} + \cos \delta_{max} = 1.114$$

[6]

Neglecting resistance the power-angle characteristics for a cylindrical-rotor synchronous generator is

$$P = P_{max} \sin \delta$$

In steady state

$$\delta_0 = \sin^{-1} \frac{P_0}{P_{max}} = \sin^{-1}(0.4) = 23.58^o = 0.41 \text{ rad}$$

[1 mark]
After mechanical input power is increased by 20%

$$P_1 = 1.2P_0 = 1.2 \times 0.4P_{max} = 0.48P_{max}$$

[1 mark]
Accelerating area is given by

$$\int_{\delta_0}^{\delta_1} (P_1 - P_{max} \sin \delta)\, d\delta$$

Decelerating area is given by

Model Answers

$$\int_{\delta_1}^{\delta_{max}} (P_{max} \sin\delta - P_1)\, d\delta$$

Using equal-area criterion

$$\int_{\delta_0}^{\delta_1} (P_1 - P_{max} \sin\delta)\, d\delta = \int_{\delta_1}^{\delta_{max}} (P_{max} \sin\delta - P_1)\, d\delta$$

[2 marks]

$$\Rightarrow \cos\delta_0 - \cos\delta_{max} = \frac{P_1}{P_{max}}(\delta_{max} - \delta_0)$$

$$\Rightarrow 0.48 \times \delta_{max} + \cos\delta_{max} = 1.114$$

[2 marks]

A few students misinterpreted the two steady-state stability limits and/or ended up calculating the accelerating/decelerating areas incorrectly.

(ii) The power output of the synchronous generator becomes zero due to a short circuit in the transmission corridor. After the short is cleared the equivalent reactance between the generator and the infinite bus becomes 0.8 p.u. Calculate the critical clearing angle in degrees.

[6]

Under pre-fault condition

$$P_{max} = \frac{E \times V}{X} = \frac{1}{0.5} = 2 \text{ p.u.}$$
$$P_0 = 0.4 \times P_{max} = 0.4 \times 2 = 0.8 \text{ p.u.}$$

[1 mark]

$$\delta_0 = \sin^{-1}\frac{P_0}{P_{max}} = 23.58^o = 0.412 \text{ rad}$$

During fault $P = 0$

Under post-fault condition, the steady-state stability limit is

$$P_{max1} = \frac{1}{0.8} = 1.25 \text{ p.u.}$$
$$\Rightarrow \delta_1 = \sin^{-1}\frac{P_0}{P_{max1}} = \sin^{-1}\left(\frac{0.8}{1.25}\right) = 0.694 \text{ rad}$$

[1 mark]

Suppose δ_{cr} is the critical clearing angle

Accelerating area is given by

Model Answers

$$\int_{\delta_0}^{\delta_{cr}} (P_0 - 0)\, d\delta = P_0(\delta_{cr} - \delta_0)$$

[1 mark]

Maximum available decelerating area is given by

$$\int_{\delta_{cr}}^{\pi-\delta_1} (P_{max1} \sin\delta - P_0)\, d\delta = P_{max1}(\cos\delta_{cr} + \cos\delta_1) - P_0(\pi - \delta_1 - \delta_{cr})$$

[1 mark]

The critical clearing angle is given by:

$$P_0(\delta_{cr} - \delta_0) = P_{max1}(\cos\delta_{cr} + \cos\delta_1) - P_0(\pi - \delta_1 - \delta_{cr})$$

$$\Rightarrow \delta_{cr} = \cos^{-1}\left[\frac{P_0}{P_{max1}}(\pi - \delta_1 - \delta_0) - \cos\delta_1\right] = 57.72^{o}$$

[2 marks]

There were several instances where the correct approach and steps were followed but the final answer was somewhat off.

In some cases (as in part i), mistakes were made in interpreting the two steady-state stability limits and calculating the accelerating and decelerating areas.

Model Answers